Disparity Analyses Are Robust To Ancestral State Estimation Uncertainty

Caleb Scutt[1,*], Natalie Cooper[2], Gavin Thomas[1], Thomas Guillerme[1]

[1]School of Biosciences, University of Sheffield, Sheffield, UK

[2]Natural History Museum, London, UK

*Corresponding author: cnscutt1@sheffield.ac.uk

Abstract

Ancestral state estimation is a widely used method to infer unobserved traits across a phylogeny, extending analyses beyond incomplete fossil records. In studies of morphological disparity ancestral state estimation can account for ghost lineages, yet estimates are often confounded with high uncertainty. How best to deal with this uncertainty remains unclear. Using simulations, we compared pre- and post-ordination estimation methods, as well as both point estimate and probabilistic approaches for treating uncertainty, under continuous and discrete trait models and varying fossil sampling levels. The relative disparity error was calculated from the difference between our estimated disparity values and the true disparity values from the original simulations. We then extended this framework to evaluate the recovery of changes in disparity across time by simulating mass extinction events with varying intensity and selectivity. We find that pre-ordination probabilistic estimations outperform post-ordination and point estimation methods and that omission of ancestral state estimations leads to underestimates of disparity. As expected, higher fossil sampling improves estimation accuracy, although recovered disparity accuracy was robust even at low fossil sampling. Similar patterns were observed when recovering changes in disparity across mass extinction events but declined significantly when fossil sampling was low, highlighting that fossil inclusion is essential when assessing disparity in deep time. These results demonstrate that explicitly propagating uncertainty, rather than collapsing states into point estimates, is more likely to recover accurate patterns in disparity, reinforcing the need to move beyond point estimates when working with the incomplete fossil record.

Keywords: ancestral state estimation; morphological disparity; fossils; simulation; uncertainty; macroevolution

1 Introduction

2 Materials and Methods

2.1 Simulation Framework

2.1.1 Phylogenies

We simulated birth-death trees with the speciation parameter set to 1.0 (λ), and the extinction one set to 0.7 (μ) (Guillerme 2024). Fixing $\lambda = 1.0$ provides a convenient relative time-scale, since only the ratio of extinction to speciation affects the shape and balance of simulated trees. These parameters correspond to a moderate-to-high turnover regime, consistent with the upper-empirical estimates from both fossil- and molecular-based diversification studies of vertebrate groups. This level of extinction produces realistic evolutionary trees with substantial extant and extinct diversity, while maintaining a positive net diversification rate ($\lambda - \mu = 0.3$). We generated three sets of 100 birth-death trees with these parameters until the simulations reached 50, 100 and 150 living taxa. Because of the randomness of the birth-death model (e.g. simulations containing only extinct taxa), we re-ran each individual simulation until we obtained the desired number of living taxa (50, 100 or 150). This means that although the input parameters where set to $\lambda = 1$; $\mu = 0.7$ the effective parameters slightly deviates from these input values (see supplementary figure @@@).

2.1.2 Trait simulation

Continuous trait data were simulated under variations of Brownian motion (BM) and Ornstein-Uhlenbeck (OU) models (table 1). For BM, the trait data were simulated as both stationary and with a moving trend. BM was simulated with variance = 0.25. The BM + trend For OU, the data were simulated as strong, weak and with differing optimum and root states. Following Gearty et al. (2024), the alpha (α) parameter was set based on phylogenetic half life ($\frac{ln(2)}{\alpha}$) to account for differing tree heights across simulations (Cooper et al. 2016). For the strong and weak OU models α was set as 10% and 50% of the tree height respectively. We simulate 100 independent characters for each model across each phylogeny.

For each phylogeny, we simulated discrete and continuous trait data separately. Discrete trait data were simulated using across three rate categories, representing fast, medium and slow transition rates (table 1). These rates were assigned by generating equal-rate (ER) transition

probability matrices, i.e., $P_{ij} = P_{ji}$. For each category, 85 binary characters and 15 three-state characters were simulated, following the number of states from empirical data (Guillerme and Cooper 2016; Casali et al. 2023). Discrete trait evolution was simulated using a discrete-time Markov process, where the probability of state transition along each branch was proportional to the product of the transition probability and branch length.

2.2 Fossil sampling

In order to simulate different levels of fossil sampling, each tree was split into chronological 'stages' to emulate the stratigraphic nature of the fossil record. Fossil tips were sampled uniformly at random from all extinct lineages within each chronological stage. Fossil sampling was carried out at five levels, ω_0, ω_5, ω_{15}, ω_{50} and ω_{100} (where $\omega_P = P\%$ fossils sampled), across each simulated phylogeny. The corresponding trait matrices were subset to match sampled taxa, with ancestral node states excluded.

2.3 Ancestral state estimation

We ran ancestral state estimation on the down-sampled matrices using the `multi.ace` function in `dispRity`, which is a wrapper for `ace` in the `ape` package (Paradis and Schliep 2019). We used four combinations of two approaches to ancestral state estimation to evaluate each method's accuracy in recovering the true disparity signal.

2.3.1 Pre-ordination estimation

Firstly, we compared pre- versus post-ordination estimation methods. The pre-ordination estimation method for estimating ancestral discrete characters involves estimating the marginal likelihoods for each discrete state at each node, based on a Maximum-Likelihood derived estimate of the Mk transition matrix and the tree topology given (Pagel 1999; Yang 2006). Given that the transition rates were set *a priori* during simulations, the equal-rates (ER) model was used to avoid model misspecification as a confounding factor on estimation accuracy (Keating 2023).

Pre-ordination estimation of continuous characters models evolution under a Brownian motion model, whereby ancestral state values lie within a multivariate normal distribution that is estimated given the observed tip data and the shared evolutionary history between taxa (Schluter et al. 1997; Rohlf 2001). Confidence intervals were calculated from standard errors derived from the Hessian matrix at the maximum likelihood estimate. The width of confidence intervals

reflects the uncertainty in continuous ancestral estimates, which typically increases with node depth and decreases with the number of informative descendant lineages.

2.3.2 Post-ordination estimation

Post-ordination estimation involves the estimation of a node's position within a multidimensional ordinated matrix (commonly referred to as the trait space, morphospace, functional space, etc.). The generation of an ordinated matrix differed between discrete and continuous characters. For discrete characters, distance matrices were computed using the 'MORD' metric (Lloyd 2016) and ordinated using principal coordinates analyses (PCoA) (`cmdscale()` function). Continuous data were ordinated using principal components analysis (PCA) (`prcomp()` function). For both the discrete- and continuous- character derived trait spaces, the ancestral states across all PC axes were then estimated under a Brownian motion model, as detailed in section 2.3.1.

2.3.3 Uncertainty treatment

For each estimation method, we also treated uncertainty in estimates with both deterministic and probabilistic approaches. The deterministic approach involves collapsing the likelihood outputs into point estimates. For discrete characters, this involved a strict majority rule whereby the state with the highest scaled likelihood was selected at each node for each character. In scenarios where state likelihoods are equal, one state is chosen at random. We also employed a relative majority rule, whereby any state at a scaled likelihood greater than the maximum observed scaled likelihood minus the inverse number of possible states is collapsed, otherwise states are left as uncertain. For continuous characters, the deterministic approach used the median of the estimated ancestral confidence interval.

The probabilistic approach differs in that uncertainty in estimates are explicitly accounted for by sampling multiple results rather than a single point estimate. For each discrete character states at a node the states where sampled a 100 times according to their scaled likelihoods (resulting in a 100 character estimation matrices). In a scenario where there is uncertainty in a binary character for a specific node, e.g., 0.55 for state 0 and 0.45 for state 1, this distribution approach would results in a 55% probability of sampling state 0 in each matrix for the respective node (and 45% chance of sampling state 1). Continuous characters were treated similarly, with 100 matrices replicated and ancestral state estimates for each node and character sampled across a normal distribution centered on the mean estimation and spread across the estimated confidence

interval. Both procedures are implemented in the `dispRity` package using the `sample = 100` argument in the `multi.ace` function (Guillerme 2018).

2.4 Disparity metrics

To ensure fair comparison between different fossil sampling levels only the nodes present across all fossil sampling levels were retained from the ancestral state estimation outputs for each phylogeny; therefore, comparisons in disparity calculations between fossil sampling levels differed solely on the difference in number of fossil tips and the estimations of node states. Discrete and continuous pre-ordination ancestral state estimations were ordinated in the same manner as described in section 2.3.2.

When assessing overall disparity, we wanted to understand how well ancestral state estimation can recover true patterns in size and density in trait space occupancy (Guillerme et al. 2020). Size-based metrics describe the spread of points in the trait space (Guillerme et al. 2020). To explore how well the size of the trait space was recovered, we calculated the sum of quantiles. The sum of quantiles calculates the range on each PC axis for the 95% of the data and then sums these ranges across all axes. This metric has similar properties to the often used sum of ranges metric by describing the total trait space exploited (Foote 1992), but with less sensitivity to outliers and is thus less sensitive to sample size. For density, we measured the sum of variances (Ciampaglio et al. 2001; Wills 2001) and the average nearest neighbour distance (Foote 1992). The sum of variances is a very widely used metric in disparity analyses. Disparity values were calculated across each estimation and the 'true' simulation so that a relative disparity error could be calculated.

$$\text{Relative disparity error} = \frac{\text{Estimated disparity} - \text{True disparity}}{\text{True disparity}} \tag{1}$$

We calculated disparity across three different treatments of the ordinated matrices. Firstly, the untreated ordination output was used. Secondly, we min-max standardised the PC axes so that the absolute magnitude in PC scores did not influence the disparity metric scores (i.e. in this case, the length of PC1 was always set to 1). Thirdly, we only used the minimum number of dimensions that were contained across all fossil sampling levels for that specific phylogeny. For the simulations of 50 living taxa trees, this would be the first 48 dimensions, in the 100 living taxa it would be 98 dimensions etc.

2.5 Mass extinction simulations

Having established the accuracy of different ancestral state estimation methods in recovering overall trait space patterns, we next tested how well these methods capture changes in disparity across time during simulated mass extinction events. Following the framework above, 100 additional trees were simulated with again 50, 100 and 150 living taxa but this time including a mass extinction event. We simulated both random mass extinctions, where taxa were a certain number of living taxa where made extinct at a specific time of the simulation. And we simulated selective mass extinctions, similar as above but where taxa were more prone to extinction based on their trait values. Extinction severity was set with a uniform distribution between 75–95%, the range seen between Cretaceous-Paleogene (66 million years ago - mya) and the end Permian extinctions mass extinction (251 mya). For selective extinctions,

With the new set of trees (and trait data for selective extinctions), the framework from section 2.1.2 to 2.3.3 was repeated. Rather than evaluating the overall disparity estimate, we assessed how well the methods could recover the relative disparity change across the extinction event. Two bins that were equidistant based off the total tree height divided by six were sampled either side of the extinction event. Disparity was measured from the taxa contained within each of these two bins to calculate a relative disparity change across the extinction event.

$$\text{Relative change in disparity} = \frac{\text{Disparity post-extinction} - \text{Disparity pre-extinction}}{\text{Disparity pre-extinction}} \tag{2}$$

The sum of quantiles, sum of variances and average nearest neighbour distance were calculated again, as well as the average displacement (Guillerme et al. 2020) as a change in position metric.

3 Results

3.1 Overall disparity error

3.1.1 Discrete

3.2 Overall Performance of ASE Methods

4 Discussion

5 Conclusions

Acknowledgments

Author Contributions

Data Availability

Competing Interests

Figures

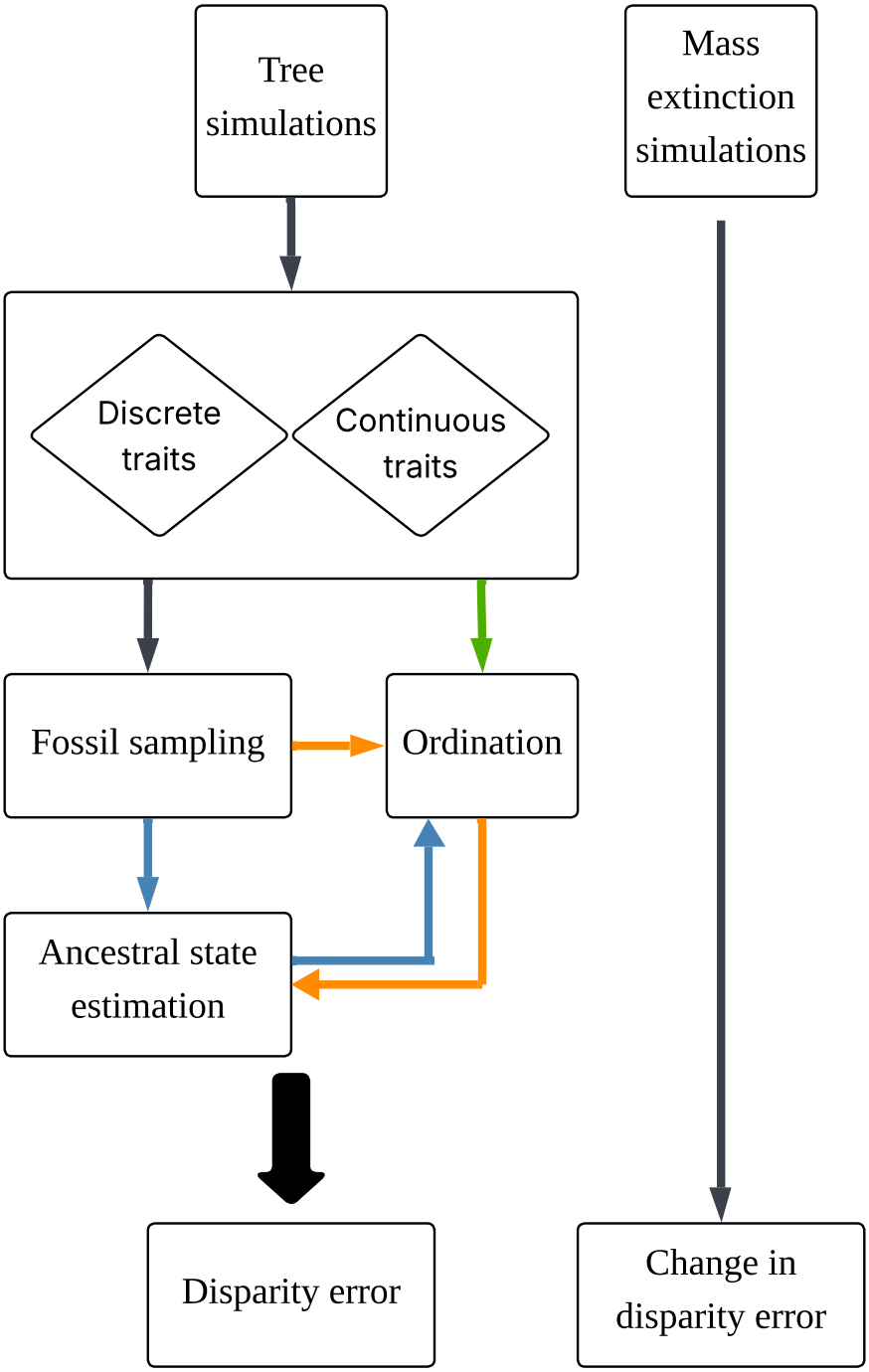

Figure 1: **Simulation workflow**
Green arrow = 'true disparity'; blue arrow = pre-ordination estimation; orange arrow = post-ordination estimation.

Tables

Table 1: Trait simulation models and parameters used across each phylogeny. For continuous characters, the root state was always set as 0. P_{ij} = transition probability; σ = variance; α = rate of adaptation toward the optimum; θ = optimum.

Type of data	**Model**	**Parameters**
Discrete	Fast	$P_{ij} = 0.9$
	Medium	$P_{ij} = 0.1$
	Slow	$P_{ij} = 0.01$
Continuous	BM	$\sigma = 0.25$
	BM (trend)	$\sigma = 0.25,\ \text{trend} = 0.3$
	OU (strong)	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height / 10}}$
	OU (weak)	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height / 5}}$
	OU (shift optimum)	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height / 5}},\ \theta = 2$

References

Daniel M Casali, Alberto Boscaini, Timothy J Gaudin, and Fernando A Perini. Morphological disparity and evolutionary rates of cranial and postcranial characters in sloths (mammalia, pilosa, folivora). *Palaeontology*, 66(1):e12639, 2023.

Charles N Ciampaglio, Matthieu Kemp, and Daniel W McShea. Detecting changes in morphospace occupation patterns in the fossil record: characterization and analysis of measures of disparity. *Paleobiology*, 27(4):695–715, 2001.

Natalie Cooper, Gavin H Thomas, Chris Venditti, Andrew Meade, and Rob P Freckleton. A cautionary note on the use of ornstein uhlenbeck models in macroevolutionary studies. *Biological journal of the Linnean Society*, 118(1):64–77, 2016.

Mike Foote. Rarefaction analysis of morphological and taxonomic diversity. *Paleobiology*, 18(1): 1–16, 1992.

William Gearty, Bethany Allen, Pedro L Godoy, and Alfio Alessandro Chiarenza. The impact of tip age distribution on reconstructing trait evolution using phylogenetic comparative methods. 2024.

Thomas Guillerme. disprity: a modular r package for measuring disparity. *Methods in Ecology and Evolution*, 9(7):1755–1763, 2018.

Thomas Guillerme. treats: A modular r package for simulating trees and traits. *Methods in Ecology and Evolution*, 15(4):647–656, 2024.

Thomas Guillerme and Natalie Cooper. Effects of missing data on topological inference using a total evidence approach. *Molecular phylogenetics and evolution*, 94:146–158, 2016.

Thomas Guillerme, Mark N Puttick, Ariel E Marcy, and Vera Weisbecker. Shifting spaces: Which disparity or dissimilarity measurement best summarize occupancy in multidimensional spaces? *Ecology and evolution*, 10(14):7261–7275, 2020.

Joseph N Keating. What is the best method for estimating ancestral states from discrete characters? *bioRxiv*, pages 2023–08, 2023.

Graeme T Lloyd. Estimating morphological diversity and tempo with discrete character-taxon matrices: implementation, challenges, progress, and future directions. *Biological Journal of the Linnean Society*, 118(1):131–151, 2016.

Mark Pagel. The maximum likelihood approach to reconstructing ancestral character states of discrete characters on phylogenies. *Systematic biology*, 48(3):612–622, 1999.

Emmanuel Paradis and Klaus Schliep. ape 5.0: an environment for modern phylogenetics and evolutionary analyses in r. *Bioinformatics*, 35(3):526–528, 2019.

F James Rohlf. Comparative methods for the analysis of continuous variables: geometric interpretations. *Evolution*, 55(11):2143–2160, 2001.

Dolph Schluter, Trevor Price, Arne Ø Mooers, and Donald Ludwig. Likelihood of ancestor states in adaptive radiation. *Evolution*, 51(6):1699–1711, 1997.

Matthew A Wills. Morphological disparity: a primer. pages 55–144, 2001.

Ziheng Yang. *Computational molecular evolution*. OUP Oxford, 2006.